



Nabil Mosharraf

Software engineer with 11+ years of cross-domain expertise in mobile, backend, and AI, focused on building and optimising production-scale products serving 2M+ users monthly.

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[LinkedIn](#)
[Github](#) (500 stars)
[Medium](#) (100k reads)
[Youtube](#)
[Substack](#)

KEY AREAS OF EXPERTISE

- Mobile Architecture: Native Android (Java/Kotlin), Flutter, Performance optimization
- Full-Stack Engineering: React.js frontend, Node.js & Golang backend, Scalable solutions
- AI & Machine Learning: Speech recognition, NLP, Vision, LLM integration, ML optimization, Data Analysis
- Leadership: Non-profit founder, Technical team leadership, Cross-functional collaboration
- Product Development: Data-driven feature prioritization, User feedback loops, Product metrics analysis

EXPERIENCE

Ikhlas.com by AirAsia – Tech Lead

September 2025 - Present | Malaysia

- Focused on optimising the codebase and remove tech debt
- Led architecture redesign of the home feed, moving from REST to GraphQL with Stellate CDN, reducing data over-fetching by 30%
- Optimised the feature flagging system, reducing tech debt and release fragileness
- Mentored 2 junior developers, conducting weekly code reviews and architecture workshops
- Optimised cloud GCP costs and reduce monthly costs by 50%.

Jobstreet by SEEK – Senior Software Engineer

July 2023 - March 2025 | Malaysia (Australia HQ)

- Worked with a team of 30 developers from Australia, Hong Kong, Philippines, and Malaysia, scaling the app to 2M MAU
- Used Kotlin, Jetpack Compose, and GraphQL to enhance app's user experience with extensive test coverage of more than 70%
- Improved the performance of Main Screen Loading by 25%.
- Won an AI Trailblazer Award for AI Interview Prep tool
- Also worked on the Graph Backend using nodejs and TypeScript

SKILLS

Language

Java (10+ years)
Kotlin (8+ years)
Dart (7+ years)
Python (4+ years)
JavaScript (4 years)

Framework

Android (10+ years)
Flutter (7+ years)
iOS (1 year)
React.js (4 years)

Backend

GoLang (2 years)
Nodejs (4 years)

Other Technologies

GraphQL, Elastic Search, Next.js, NLP (Huggingface), Appium, Jenkins

NOTABLE PORTFOLIO & OPEN SOURCE PROJECTS

- **Co-Founder of gtaf.org**, a non profit organisation making islamic apps accessible to people
- **Al Quran (Tafsir & by Word)**: Reached Top 5 in Free Books category in multiple countries, 50M+ Downloads with 2M MAU
- **Jobstreet Android**: Major job platform application scaling to 1.2M MAU
- **GraphView Library**: 470+ stars, 11.9k+ downloads – Graph visualization library for Flutter | [Github](#)
- **FlutterDownloadManager**: 69+ stars, 4.63k+ downloads – Cross-platform download manager | [Github](#)
- **LottieSwipeRefreshLayout**: Custom SwipeRefreshLayout with Lottie

Photobook Worldwide Sdn Bhd — *Lead Android Developer*

Nov 2019 - July 2023 | Malaysia

- Leading an agile team of 5 Android Developers.
- Developed new editor foundation enabling 6 additional photo editor types and expanding product options
- Optimized with Kotlin coroutines: 50% faster photo loading and improved face detection accuracy
- Implemented CI/CD with Jenkins automating 90+ daily UI tests, reducing regression bugs by 75%
- Spearheaded Jetpack Compose adoption for dynamic home screen, cutting development time from 3 to 1.5 months

MYCASH ONLINE (SG) PTE LTD — *Senior Android Developer*

Jan 2018 - Aug 2018 | Malaysia

- Reduced app size from 21mb to 4mb using proguard and other optimisation techniques
- Integrated payment gateway using MOL payment gateway (Malaysia) and Poli payment Gateway (Australia)
- Used Databinding using MVVM Pattern
- Wrote maintainable and reusable Java/kotlin code with a focus on security and privacy

Greentech Apps Foundation — *Co-Founder and CTO, later R&D Lead*

Jan 2015 - Oct 2019 (Full-time CTO) ; Nov 2022 - June 2024 (Part-time R&D Lead) | UK (Remote); Currently Board Member

Executive Leadership

- Managed company budget and built an executive team (Heads of Eng, Marketing, Ops).
- Led distributed, cross-functional team of 25+ (15 engineers) across multiple cities.
- Designed a 5-point performance evaluation system and targeted improvement plans.

Product Growth

- Scaled platform to 50M+ downloads and 2M+ MAU as founding developer; flagship app hit Top 5 in Free Books globally.
- Launched 5+ Islamic apps from concept to release, reaching 15M+ collective downloads.

Engineering & AI

- Spearheaded AI/R&D: fine-tuned wav2vec2.0 for Arabic mispronunciation detection, built audio alignment algorithms, and integrated open-source LLMs for support.
- Optimized app load times to <1 sec, reducing bounce rates by 30%.
- Engineered C++ NDK solution for Arabic text rendering on legacy devices.
- Built Go/Swift OPUS-to-CAF converter, optimizing Apple ecosystem

EDUCATION

University Malaya — *MSc Computer Science*

2019 - 2022

Masters in Computer Science
(Applied Computing)

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH — *BSc in Electrical and Electronics Engineering*

2011- 2014

Major in EEE scoring a CGPA of 3.94 out of 4 (Magna Cum Laude holder)

LANGUAGES

English (IELTS: 7.5)

Bengali

bandwidth.

- Deployed cross-platform Flutter applications (Mobile & Desktop).

Operations & Process

- Implemented Agile workflows and man-week planning, improving sprint predictability by 1-2 weeks.
- Instituted strategic code freezes and roadmap planning for peak seasonal demand (Ramadan).
- Merged Product and Design functions to streamline handoffs and organizational efficiency.

PUBLICATIONS

- The 30% Wardrobe Hypothesis: Building a Personal Clothes Tracking System ([Link](#))
- Optimizing my Opus to CAF Converter in Go: A 40% Performance Boost ([Link](#))
- Creating Audiobooks by aligning Audio with text ([Link](#))
- Text similarity algorithms for matching Islamic Hadith Documents for Hadith Interlinking. ([Link](#))
- Avoid CORS requests for a React App ([Link](#))
- Comparison of string encryption performance using symmetric methods in Android ([Link](#))
- A Novel LLM-Based Approach for Automated Seerah-Hadith Mapping: Connecting Islamic Historical Narratives Through Vector Search and Semantic Analysis ([Link](#))
- ADAB: A Culturally-Aligned Automated Response Generation Framework for Islamic App Reviews by Integrating ABSA and Hybrid RAG ([Link](#))
- A Comparative Study of Pretrained Transformer Models for Quranic ASR: Speech Representations, Label Formats, and Dataset Composition (Coming Soon)